SIMATS ENGINEERING

CSA0705 - COMPUTER NETWORKS

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1. Introduction and Objective

The plant in this assignment is a smart manufacturing plant. It has a robot cell next to welding machines, automated guided vehicles (AGVs) that move around the floor, CCTV cameras, vibration and other low-data-rate sensors, and office users. A shop-floor switch room and a data centre support these, and the head office is 400 km away.

The network must be reliable, low in delay and able to grow. Robot commands have the strictest requirement, a delay of 10 ms (Task 1). CCTV creates the largest traffic volume: one production line needs 158.24 Mbps and eight lines need 1,265.92 Mbps. A weak design would either miss the robot deadline or run out of capacity.

The objective of this report is to defend one final network design using the results of Tasks 1 to 9. It answers the four Task 11 questions: the medium and settings for each area (Sections 2 and 3), the error-control and flow-control choices (Section 4), the two biggest remaining risks (Section 5), and the five-plant question (Section 6). Section 7 answers the private 5G alternative concept.

Data sources and assumptions. Numerical values from Tasks 1, 4, 5, 8 and 9 are used exactly as calculated in those tasks. They are calculated or scenario values, not measurements taken on a live plant. No numerical result from Task 10 was available while this report was prepared, so none is quoted. Cable-length limits (for example 100 m for a twisted-pair channel) are standard values. Performance figures for Wi-Fi 6 and private 5G are design targets and must be confirmed by testing.

2. Recommended Network Architecture

Table 1 gives the recommended medium, technology and settings for each area, with the reason for each choice.

Area	Recommended medium / technology	Ethernet or wireless configuration	Technical justification
1. Robot cell near welding machines – 85 m	Cat6A S/FTP shielded copper	1000BASE-T, full duplex (10GBASE-T possible). Auto-negotiation on; both ends agree speed and duplex. MTU 1500. Priority queue (IEEE 802.1p / DSCP) for robot-control traffic.	85 m is inside the 100 m twisted-pair channel limit (patch cords must be included in the 100 m). Shielded pairs resist welding noise. Wired delivery avoids the 15 dB SNR problem found in Task 4 and gives steady delay for the 10 ms deadline. If testing shows errors, replace with fibre.
2. Office to shop-floor switch – 140 m	Multimode fibre (OM3 or OM4)	10GBASE-SR (or 1000BASE-SX), full duplex. Fibre ports use a fixed speed set by the transceiver; both ends must use the same type. MTU 1500; VLAN trunk.	140 m is beyond the 100 m copper limit. OM3 supports 10GBASE-SR up to about 300 m and OM4 up to about 400 m, so 140 m is well within reach. Fibre is not affected by electrical noise.
3. Shop floor to data centre – 550 m	Single-mode fibre (OS2)	10GBASE-LR, full duplex, reach up to 10 km. Two fibres in a link-aggregation pair, preferably on separate routes. MTU 1500.	550 m is longer than the multimode 10 Gbps reach (300 m on OM3, 400 m on OM4), so single-mode is needed. It carries all plant traffic, so it has the most spare capacity and a backup path.
4. Plant to head office – 400 km	Leased fibre / carrier Ethernet service (operator-managed)	Full-duplex Ethernet hand-off, starting at 10 Gbps with a service-level agreement. MTU 1500 end to end (confirm with the carrier). Router and firewall with an encrypted VPN; QoS markings agreed with the carrier.	400 km is far beyond unamplified fibre reach, so the operator's transport network is required. One-way propagation is 2 ms (Task 9). Capacity is justified in Section 6.
5. Moving AGVs	Private 5G with URLLC	Wireless, so there is no Ethernet duplex setting on the radio link. Radio units are wired to the core switch by full-duplex Ethernet or fibre. Overlapping cell coverage for handover; separate QoS class for AGV control.	AGVs move at 2 m/s. Handover takes 20 ms on private 5G and 150 ms on Wi-Fi (Task 5). The AGV moves 0.04 m during a 5G handover and 0.30 m during a Wi-Fi handover.
6. Office users	Wi-Fi 6 (IEEE 802.11ax)	Wireless; access points are wired with full-duplex Ethernet and PoE. Separate office VLAN and SSID, WPA3 security, planned channels.	Office users are fixed or slow-moving. ERP and web traffic accept some delay, so the 150 ms Wi-Fi handover is acceptable. Wi-Fi 6 gives high throughput at lower cost than 5G.
7. CCTV cameras	Wired Ethernet: PoE Cat6/Cat6A (up to 100 m) with fibre uplinks	1000BASE-T, full duplex, auto-negotiation on. Separate CCTV VLAN in a lower QoS class, with local recording.	CCTV is the largest traffic source: 158.24 Mbps per line with continuous video, or 67.04 Mbps per line if only 5% is sent continuously (Task 1). Keeping it wired protects the wireless capacity.
8. Vibration and other low-data-rate sensors	Wired Ethernet or fieldbus where cabling is easy; LoRaWAN or Zigbee gateway for hard-to-cable positions	Wired: 100 Mbps full duplex is enough. Wireless: low-power gateway connected to a switch by Ethernet.	Readings are small and periodic, so they need little capacity. Low-power radio saves cabling. Sensors that send raw high-rate waveforms should stay wired.

Table 1: Recommended medium, technology and settings for each area.

Ethernet settings used throughout

- Full-duplex operation on every wired link, so collisions do not occur.
- Auto-negotiation enabled on compatible copper links; both ends must end up with the same speed and duplex, because a mismatch causes errors and slow throughput. Fibre links are fixed by the transceiver type.

- Consistent MTU (1500 bytes, the frame size used in Task 9) along the whole path, so frames are not dropped or fragmented.
- QoS prioritisation: robot-control traffic in the highest queue, AGV and sensor alarms next, office/ERP traffic after that, and CCTV in a rate-limited lower class.
- Wi-Fi 6 and 5G are wireless. They do not use Ethernet duplex settings; only their wired uplinks do.

Why the design is hybrid

A fully wired plant cannot serve AGVs, which must move. A fully wireless plant would put the heavy CCTV load and the 10 ms robot traffic on shared radio, where interference and handover add delay and loss (Section 5). The hybrid design uses cable where devices are fixed and traffic is heavy or time-critical, and wireless only where mobility or convenience is needed: private 5G for AGVs and Wi-Fi 6 for office users.

3. Network Architecture Diagram

Figure 1 shows how the parts connect. Solid lines are wired links and dashed lines are wireless links. The five-plant concept is shown at the top: every plant uses the same design and connects to the head office over the WAN.

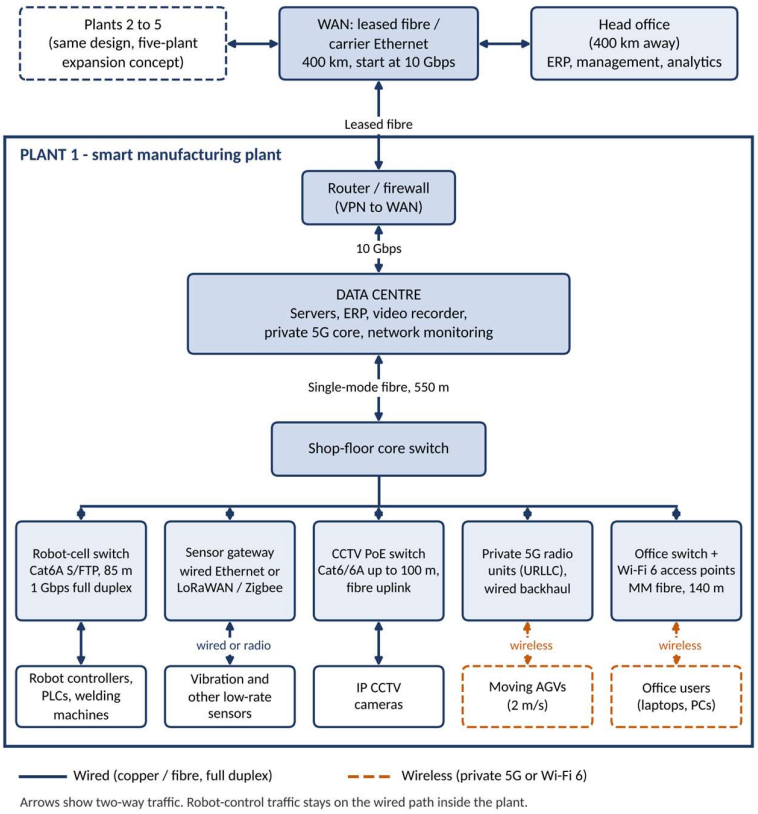


Figure 1: Proposed hybrid network architecture for Plant 1 and the five-plant expansion concept.

4. Error-Control and Flow-Control Decisions

Four terms must be kept separate. CRC detects a corrupted frame but does not correct it. FEC adds extra bits so the receiver can correct some errors without asking for a resend. ARQ resends data that was lost or corrupted. Flow control stops a fast sender from overwhelming a slow receiver; it is different from congestion control, which protects the network. Different links use different methods, as Table 2 shows.

Traffic / link	Recommended error-control method	Flow-control or reliability method	Reason and calculation from earlier tasks
Robot-control traffic	CRC-32 (Ethernet frame check sequence) detects corrupted frames, which are discarded. Where a radio or other link supports FEC, FEC corrects small errors without waiting. No reliance on retransmission.	Real-time protocol with cyclic updates, so the next command replaces a lost one. A watchdog timer puts the robot in a safe state if updates stop. Highest QoS queue. No TCP.	Retransmission delay is 40 ms (Task 8), which is 4 times the 10 ms deadline, so waiting for a resend cannot meet it. Hamming(21,16) shows how FEC works: 5 check bits, 31.25% overhead, corrects 1 bit. It is an example, not the best method for all robot traffic.
Vibration-sensor readings	CRC detects errors. LoRa radio includes its own FEC. Critical alarms use an acknowledgement and retry (ARQ).	Periodic readings need no window; a lost reading is replaced by the next one. Alarms use Stop-and-Wait, which is acceptable for small messages.	Data rate is low and links are short. Stop-and-Wait is inefficient only when a is large (a = 166.67 on the WAN, Task 9), which is not the case on a local sensor link.
CCTV traffic	CRC detects damaged frames. Live video is not retransmitted because late frames are useless; the decoder hides small losses. Local recording keeps complete footage.	Encoder bit rate set by the camera, plus QoS rate limiting. Live streaming usually uses UDP/RTP; recorded files copied later use TCP.	Task 1: 158.24 Mbps per line with continuous video; 67.04 Mbps per line with 5% sent. The main control needed is limiting volume, not correcting every error.
Office / ERP traffic	Ethernet or Wi-Fi frame check sequence (CRC) plus TCP checksum. TCP resends lost data (ARQ).	TCP sliding-window flow control using the receiver's advertised window. TCP congestion control is a separate mechanism.	ERP needs complete and correct data. A 40 ms retransmission (Task 8) is harmless for office applications, so ARQ is the right choice here.
Head-office WAN (400 km)	CRC on each Ethernet hop detects errors. TCP with selective acknowledgement resends only the lost segments.	Sliding window of at least 335 frames (about 0.5 MB). Stop-and-Wait is not suitable.	Stop-and-Wait utilisation is only 0.299% (Task 9). Selective Repeat resends one frame after an isolated loss, while Go-Back-N may resend up to 335 frames.

Table 2: Error-control and flow-control choices for each traffic type and link.

4.1 Hamming code calculation (Task 8)

For m data bits, r check bits must satisfy the following condition. For m = 16:

$$2^{r} \geq m + r + 1$$

With r = 4, $2^4 = 16$ is less than 16 + 4 + 1 = 21, so 4 bits are not enough. With r = 5, $2^5 = 32$ is at least 16 + 5 + 1 = 22, so 5 check bits are needed. The transmitted word is 16 + 5 = 21 bits. The overhead relative to the original data is:

$$\text{Overhead} = (5 \div 16) \times 100\% = 31.25\%$$

A Hamming distance of 3 can detect 2-bit errors or correct 1-bit errors. It cannot correct burst errors, so FEC supports good cabling and shielding but does not replace them. For robot traffic the retransmission delay of 40 ms is 4 times the 10 ms limit (40 ms ÷ 10 ms = 4), which is why waiting for a resend is not suitable there.

4.2 Head-office WAN calculation (Task 9)

Given: distance d = 400 km, link speed R = 1 Gbps, frame size L = 1,500 bytes = 12,000 bits, signal speed v = 2×10^8 m/s in fibre.

$$t_t = L \div R = 12{,}000 \text{ bits} \div 10^9 \text{ bits/s} = 12 \text{ µs}$$

$$t_p = d \div v = 400{,}000 \text{ m} \div (2 \times 10^8 \text{ m/s}) = 2 \text{ ms}$$

$$a = t_p \div t_t = 2{,}000 \text{ µs} \div 12 \text{ µs} = 166.67$$

$$U_{SW} = 1 \div (1 + 2a) = 1 \div 334.33 \approx 0.299\%$$

$$W_{min} = 1 + 2a = 334.33 \approx 335 \text{ frames}$$

At 0.299% utilisation, Stop-and-Wait would carry only about 0.299% × 1,000 Mbps ≈ 2.99 Mbps on this 1 Gbps link, so a sliding window is required. Sequence numbers must cover the window:

- **Go-Back-N:** window ≤ $2^n - 1$. With 9 bits the limit is 511, which is at least 335; with 8 bits it is only 255. So 9 bits are needed.
- **Selective Repeat:** window ≤ 2^{n-1}. With 10 bits the limit is 512, which is at least 335; with 9 bits it is only 256. So 10 bits are needed.

After one isolated loss, Selective Repeat resends one frame (12,000 bits), while Go-Back-N may resend up to 335 frames (about 4.02 Mbit). The price is receiver buffering: up to 335 frames × 1,500 bytes ≈ 502,500 bytes (about 0.5 MB), which is small for a router. In practice TCP with selective acknowledgement behaves like Selective Repeat. The round-trip time is 4 ms, so the window must hold 1 Gbps × 4 ms = 4,000,000 bits = 500,000 bytes. A default 64 KB TCP window would limit throughput to about 65,535 × 8 ÷ 0.004 s ≈ 131 Mbps, so TCP window scaling must be enabled on the WAN routers and servers.

5. Two Biggest Remaining Risks

Risk	Why it matters	Supporting task result	Mitigation
Risk 1: Network congestion	If uplinks fill up, queues grow and frames are dropped. Robot-control packets could be delayed beyond the 10 ms limit.	Task 1: one line = 158.24 Mbps; eight lines = 1,265.92 Mbps. A single 1 Gbps uplink is insufficient: 1,265.92 Mbps is 126.6% of 1,000 Mbps, a shortfall of 265.92 Mbps. CCTV is the largest traffic source; robot commands have the strictest delay (10 ms).	Record CCTV locally and send only 5%: 536.32 Mbps for eight lines, which is 53.6% of 1 Gbps. Strict-priority QoS for robot control; separate VLANs. 10 Gbps uplinks or link aggregation; traffic monitoring and capacity planning. This reduces the risk but does not remove it, because peak bursts may exceed the averages.
Risk 2: Wireless interference and handover	Welding arcs and metal surfaces disturb radio signals. A moving AGV changes cells, and the link is briefly interrupted.	Task 4: welding-area SNR is 15 dB (a power ratio of about 31.6). Task 5: Wi-Fi handover = 150 ms, private 5G handover = 20 ms. At 2 m/s, the AGV moves 2 × 0.15 = 0.30 m (Wi-Fi) or 2 × 0.02 = 0.04 m (5G) during handover.	Radio site survey and measurements in the welding area before installation. Overlapping 5G coverage with redundant radio units. AGV stops or slows safely on its own if the link is lost; safety does not depend on the network. Keep robot control wired; monitor signal quality, handover failures and latency. Interference and handover cannot be fully eliminated; they must be tested.

Table 3: The two biggest risks that remain in the design.

The 20 ms handover on private 5G is longer than the 10 ms robot-command limit. This is one more reason why robot control stays on Cat6A cable and the AGVs rely on onboard safety logic during a handover.

6. Scalability for Five Plants

The traffic figures come from Task 1. The total is the traffic per line, multiplied by the number of lines, multiplied by the number of plants:

$$T_{\text{total}} = T_{\text{line}} \times 8 \text{ lines} \times 5 \text{ plants}$$

Scenario	One line	Eight lines (one plant)	Five plants	Share of a 10 Gbps link
Continuous CCTV	158.24 Mbps	158.24 × 8 = 1,265.92 Mbps	1,265.92 × 5 = 6,329.6 Mbps ≈ 6.33 Gbps	63.3%
Local CCTV recording, 5% of continuous video sent	67.04 Mbps	67.04 × 8 = 536.32 Mbps	536.32 × 5 = 2,681.6 Mbps ≈ 2.68 Gbps	26.8%

Table 4: Calculated average traffic (Task 1) for one plant and for five plants.

Does the design still work for five plants? Yes, with one change: the WAN capacity must be increased. The plant-level design can be copied to each plant, because robot and AGV control stay inside each plant. This is necessary, because a 400 km link already uses 2 ms one way, so a round trip of 4 ms would use 40% of the 10 ms robot deadline before any processing.

- A single 1 Gbps head-office connection is insufficient: even one plant with continuous CCTV needs 1,265.92 Mbps, and five plants need about 6.33 Gbps.
- A 10 Gbps backbone/WAN capacity is a reasonable starting recommendation for the calculated average traffic. With continuous CCTV, the average uses 63.3% of 10 Gbps, which leaves little spare room, so local recording (26.8%) is strongly preferred.
- These are average values. Actual peak traffic, protocol overhead, growth, redundancy and service-level requirements must be checked before final procurement. A second route or a second link gives protection if one fails.
- Local recording, QoS, traffic monitoring and capacity planning are needed at every plant.

7. Alternative Concept: Entire Plant Uses Private 5G

Private 5G URLLC is suitable for mobile AGVs and selected wireless industrial control, subject to measured latency, reliability and coverage. It does not guarantee zero delay or zero packet loss. Using the Task 1 figures, the split is shown in Table 5.

Traffic type	Should use private 5G?	Should remain wired?	Reason
Robot commands	No (not as the main path)	Yes	The 10 ms deadline (Task 1) leaves little margin. Wired Cat6A gives steady delay. 5G handover is 20 ms (Task 5). 5G can be considered later for selected mobile robots, after measurement.
AGVs	Yes	Only the radio units' backhaul	AGVs move at 2 m/s. 5G handover is 20 ms against 150 ms for Wi-Fi (Task 5), which means 0.04 m of travel instead of 0.30 m.
CCTV	No (only a few mobile or hard-to-wire cameras)	Yes	CCTV is the largest traffic: 1,265.92 Mbps for eight lines. This load would use much of the shared radio capacity of a cell, which typically lies in the range of hundreds of Mbps (an assumption to be confirmed with the vendor).
Office users	Not necessary	Wired uplinks; Wi-Fi 6 for devices	Users are not mobile at speed, and Wi-Fi 6 is cheaper. 5G SIMs and devices add cost.
Vibration sensors	Optional	Wired where practical	Data rate is very low. LoRaWAN or Zigbee gateways use less power and cost less. 5G suits only sensors that need a stricter delay.
Data-centre / backbone traffic	No	Yes	Fixed, very high-volume links between switches and servers. Fibre gives 10 Gbps or more and the same link also carries the 5G backhaul.

Table 5: Traffic split for the private 5G alternative concept.

Criterion	Wired Ethernet / fibre	Wi-Fi 6	Private 5G (URLLC)
Throughput	1 to 10 Gbps per link, not shared with other links	High peak rate, shared by all users of an access point	Shared by all devices in a cell; usually lower than wired
Delay	Lowest and most predictable	Varies with load and interference; handover 150 ms (Task 5)	Low target delay; handover 20 ms (Task 5)
Mobility	None	Limited (walking users)	Designed for moving equipment
Reliability	Highest when installed and shielded correctly	Affected by interference and contention	Good with planned coverage, but not guaranteed
Cost	Cabling work is costly, running cost is low	Low	Highest: core, radio units, spectrum, SIMs

Table 6: Comparison of the three technologies. Values are design-level; actual figures must be measured.

Final decision. A hybrid wired-fibre + Ethernet + Wi-Fi 6 + private 5G architecture is preferable to replacing the entire plant network with private 5G. Fibre stays as the backbone, cable carries robot control and CCTV, Wi-Fi 6 serves office users, and private 5G carries only the traffic that needs mobility with low delay: the AGVs.

8. Final Recommendation and Conclusion

The recommended design is hybrid. Cat6A S/FTP serves the 85 m robot cell, multimode fibre the 140 m office link, and single-mode fibre the 550 m link to the data centre. A leased fibre / carrier Ethernet service connects the plant to the head office. Private 5G with URLLC serves moving AGVs, Wi-Fi 6 serves office users, wired Ethernet and fibre carry CCTV, and wired or low-power links serve the sensors. All wired links use full duplex, consistent MTU and QoS.

Key figures used in this report:

- **Task 1:** 158.24 Mbps per line; 1,265.92 Mbps for eight lines; about 6.33 Gbps for five plants (or about 2.68 Gbps with local CCTV recording); robot deadline 10 ms.
- **Task 5:** handover 20 ms (private 5G) against 150 ms (Wi-Fi) at 2 m/s, which is 0.04 m against 0.30 m of AGV travel.
- **Task 8:** Hamming(21,16) uses 5 check bits, with 31.25% overhead, and corrects 1 bit; retransmission delay 40 ms.
- **Task 9:** a = 166.67; Stop-and-Wait utilisation 0.299%; minimum window 335 frames; 9 sequence-number bits for Go-Back-N and 10 for Selective Repeat.

The design meets the robot deadline by keeping robot control on cable, avoids waiting for retransmissions where time matters, handles CCTV by wiring it and recording locally, supports sensors with low-cost links, and gives the AGVs a mobile link. It scales to five plants if the WAN is upgraded to at least 10 Gbps. Two risks remain: congestion, and wireless interference and handover. They are reduced, not eliminated. The values in this report are calculated, so actual deployment requires site surveys, testing and continuous monitoring before the design is accepted.